

C Programming - Assignment

Q.1 Implement Priority Queue Using Array (Min Priority Queue)

Problem Statement

Write a program to implement a **Min Priority Queue** using an array.

In a **Min Priority Queue**, the element having the **lowest priority value** is served (deleted) first.

The program should support the following operations:

- Insert
- Delete
- Peek
- Display

Concepts Used

- Priority Queue
- Arrays
- Structures
- Functions
- Looping (**for**)
- Conditional Statements (**if**)
- Menu-Driven Programming

Step-by-Step Approach

Insert

1. Check whether the queue is full.
2. Accept the data and its priority.
3. Insert the element at the rear of the queue.
4. Increment the rear pointer.

Delete

1. Check whether the queue is empty.
2. Find the element having the **minimum priority value**.
3. Display the deleted element.
4. Shift the remaining elements to fill the gap.
5. Decrement the rear pointer.

Peek

1. Check whether the queue is empty.
2. Find the element having the minimum priority.
3. Display the element without deleting it.

Display

1. Check whether the queue is empty.
2. Traverse the queue from front to rear.
3. Display the data along with its priority.

Test Cases

Test Case 1 – Insert

Input

```
Insert (10, Priority = 3)
Insert (20, Priority = 1)
Insert (30, Priority = 5)
Insert (40, Priority = 2)
```

Output

Data	Priority
10	3
20	1
30	5
40	2

Test Case 2 – Delete

Before Delete

Data	Priority
10	3
20	1
30	5
40	2

Output

```
Deleted Element = 20
```

After Delete

Data	Priority
10	3
30	5
40	2

Test Case 3 – Peek**Input**

Current Queue

10 Priority = 3
30 Priority = 5
40 Priority = 2

Output

Front Element = 40
Priority = 2

Test Case 4 – Queue Underflow**Input**

Delete from an Empty Queue

Output

Queue Underflow

Test Case 5 – Queue Overflow**Input**

Insert elements until the queue becomes full.

Insert one more element.

Output

Queue Overflow

SUNBEAM